# Project Proposal: Understanding the Relationship Between Parameter Updates and Privacy Leakage in LLM Fine-Tuning

*Based on: Benchmarking Empirical Privacy Protection for Adaptations of Large Language Models (2026) and DP-SelFT: Differentially Private Selective Fine-Tuning for Large Language Models (2026)*

## 1. Motivation and Problem Statement

Fine-tuning Large Language Models (LLMs) on downstream data can cause information from the adaptation dataset to be memorized and subsequently leaked. Differential Privacy (DP) provides formal privacy guarantees, but the practical privacy protection of an adapted LLM can still depend on the characteristics of the adaptation process. The first paper systematically studies this issue using membership inference attacks and canary extraction, showing that the distribution of adaptation data and the choice of adaptation method can significantly affect empirical privacy leakage.

The second paper, DP-SelFT, explores *selective fine-tuning*, where only selected layers/parameters are updated during DP fine-tuning. It demonstrates that restricting where updates are applied can improve the privacy–utility trade-off.

Our project aims to connect these two ideas by directly investigating the following question:

**How does the fraction and location of trainable parameters affect empirical privacy leakage during LLM fine-tuning?**

The goal is to determine whether updating fewer parameters consistently reduces privacy leakage, and whether *which* parameters are updated is as important as *how many* parameters are updated.

## 2. Proposed Approach

We will establish a controlled fine-tuning and privacy-evaluation pipeline based on the experimental framework of the two papers. A pretrained language model will be adapted to a downstream task using different amounts of trainable parameters. We will consider several settings, ranging from updating a large fraction of the model to highly selective fine-tuning and parameter-efficient methods such as LoRA.

For a fixed task and comparable privacy setting, we will vary:

- **Parameter fraction:** percentage of model parameters that are trainable.
- **Parameter location:** early, middle, or late layers, and selected layer subsets.
- **Adaptation method:** full fine-tuning, selective fine-tuning, and parameter-efficient fine-tuning.
- **Privacy level:** different DP noise/privacy-budget settings where computationally feasible.

For every configuration, we will measure both **utility** and **privacy leakage**. Utility will be evaluated using the downstream task performance, while privacy will be evaluated using membership inference attacks and, where feasible, canary extraction following the first paper.

## 3. Experimental Analysis

The central analysis will be to construct the relationship

$$\text{Fraction of Parameters Updated} \quad \longrightarrow \quad \text{Empirical Privacy Leakage.}$$

We will compare privacy leakage against the number of trainable parameters while keeping other factors as controlled as possible. We will additionally investigate whether two models updating the same number of parameters can have different leakage depending on *which layers* are updated.

This will allow us to study three important questions:

1. Does reducing the number of updated parameters reduce empirical privacy leakage?
2. Does the location of the updated parameters influence leakage independently of the parameter count?
3. What parameter-update strategy provides the best **privacy–utility trade-off**?

## 4. Expected Contribution

Rather than reproducing either paper independently, our project will provide a controlled study of **parameter-update size and location as explicit factors governing empirical privacy leakage**. The resulting privacy–utility curves can help identify how much of an LLM actually needs to be updated for effective adaptation without unnecessarily increasing privacy risk.

If the experiments reveal consistent patterns, we will use them to propose a simple parameter-selection strategy that preferentially updates parameters providing high task utility while limiting privacy leakage. The final outcome will therefore be an empirical characterization of the relationship between **model adaptation, parameter updates, privacy leakage, and utility**.

## References


1. B. Marek et al., "Benchmarking Empirical Privacy Protection for Adaptations of Large Language Models," ICLR 2026. https://arxiv.org/abs/2606.09401
2. H. Sha et al., "DP-SelFT: Differentially Private Selective Fine-Tuning for Large Language Models," 2026. https://arxiv.org/abs/2605.17432